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constexpr tagging

pointer

This paper proposes a new library non-owning pointer and value pair which has ability to store small amount of information 'tag' in a pointer bits in an unspecified-way. This functionality must be usable in `constexpr` environment.

Revision history

- [R1](#) → [R2](#): simplifying design by removing schemas and the heavy interface, massive simplification as discussed in SG1 (`pointer_tag_pair`, `required sizeof(pointer_tag_pair) == sizeof(T*)`)
- [R0](#) → [R1](#): proposing and explaining design based on existing implementation

TL;DR

This paper proposes simple type for which user request n -bits of storage inside a pointer value in an unspecified way. Construction of the type should fail in case when no such storage is available.

Introduction and motivation

Pointer tagging is widely known and used technique ([Glasgow Haskell Compiler](#), LLVM's `PointerIntPair`, `PointerUnion`, [CPython's garbage collector](#), [Objective-C](#) / [Swift](#), Chrome's [V8 JavaScript engine](#), [GAP](#), [OCaml](#), [PBRT](#)). All major CPU vendors provides mechanism for pointer tagging (Intel's [LAM linear address masking](#), AMD's [Upper Address Ignore](#), ARM's [TBI top byte ignore](#) and [MTE memory tagging extension](#)). All widely used 64 bit platforms are [not using more than 48 or 49 bits of the pointers](#).

This functionality widely supported can't be expressed in a standard conforming way.